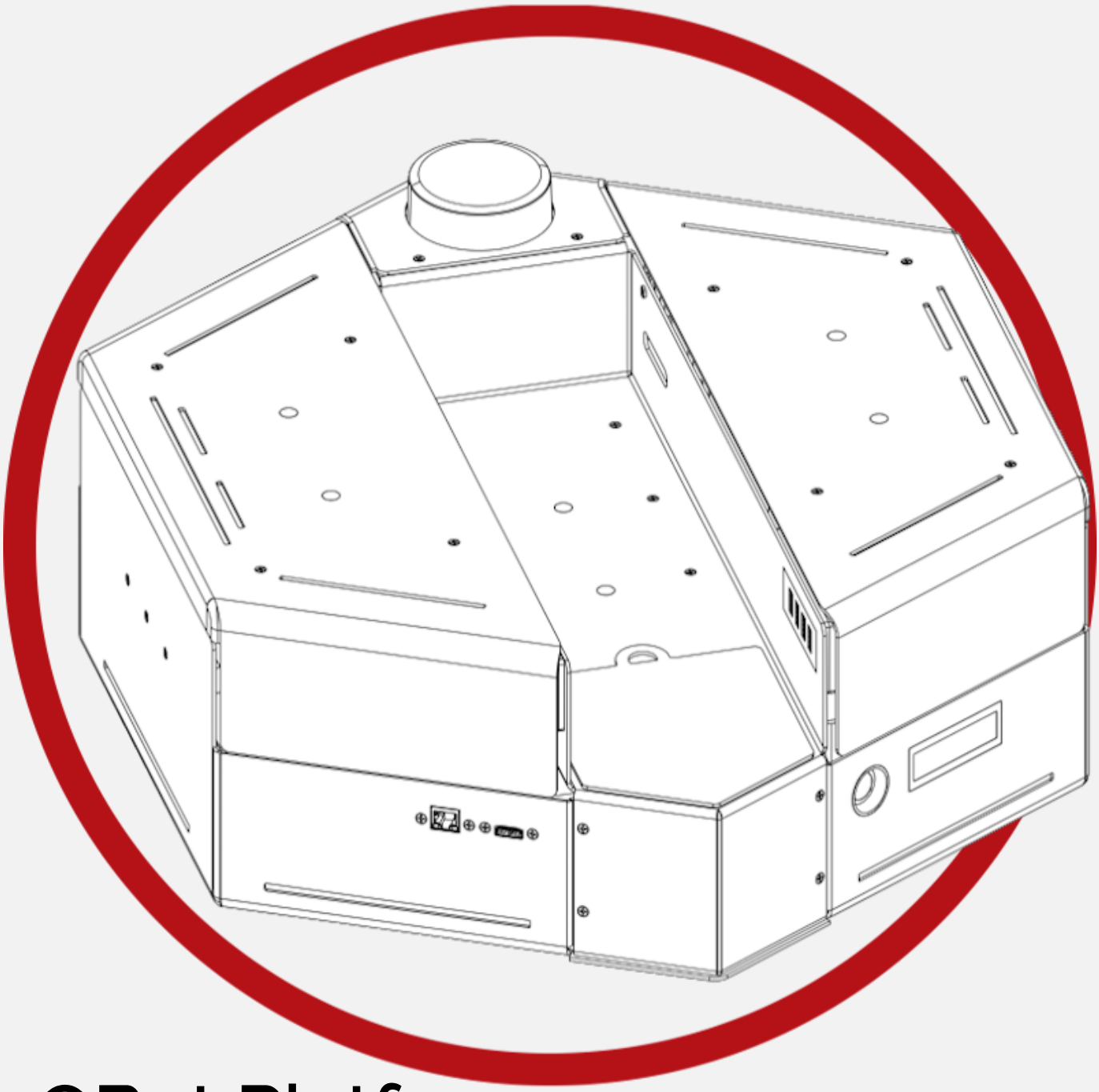
QUANSER
INNOVATE·EDUCATE



QBot Platform

Self-Localization

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Quanser Inc.
119 Spy Court
Markham, Ontario
L3R 5H6
Canada

info@quanser.com
Phone: 19059403575
Fax: 19059403576
Printed in Markham, Ontario.

For more information on the solutions Quanser Inc. offers, please visit the web site at:
http://www.quanser.com

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QBot Platform – Application Guide

Self-Localization

Why Explore Self-Localization?

Localization techniques allow autonomous robots to use information from their environment to estimate their location and orientation. LiDAR sensors are widely used for self-localization, due to its advantages in providing very accurate depth information and operating consistently in low-lighting conditions. Localization also enables robots to perform increasingly complex tasks such as navigation and path planning.

Scan Match Localization

This is the first lab of Skills Progression 2: Surveying. In this lab, the line following and obstacle detection functionality developed during Skills Progression 1 is excluded from the code for simplicity. You will use the same joystick or keyboard controls from the previous Skills Progressions to control the motion of the QBot Platform.

Code for this lab will build on some elements from the Obstacle Detection lab, which perform preprocessing on the LiDAR data by adjusting it to account for the LiDAR's forward displacement on the QBot and resampling it for fewer data points. In this lab you will use Quanser's implementation of the Scan Match algorithm to perform localization of the QBot using the preprocessed LiDAR scan data. You will explore the performance effects of tuning algorithm parameters required by Scan Match.

Note: code for this skills progression does not include implementations for obstacle avoidance. You are required to ensure the QBot does not run into obstacles while manually driving it.

Before you begin

Please review the following before beginning this lab,

1. Ensure you have completed the following labs in your language of choice:
    a. **Play** lab from Skills Progressions 0
    b. **Obstacle Detection** lab from Skills Progressions 1
2. Ensure that you have read the **Lidar Inverse Measurement Model** and **Scan Match** concept reviews.
3. If you are completing the lab with hardware, set up an environment by arranging walls surrounding the QBot with some unique features. A sample wall layout is shown in Figure 1:

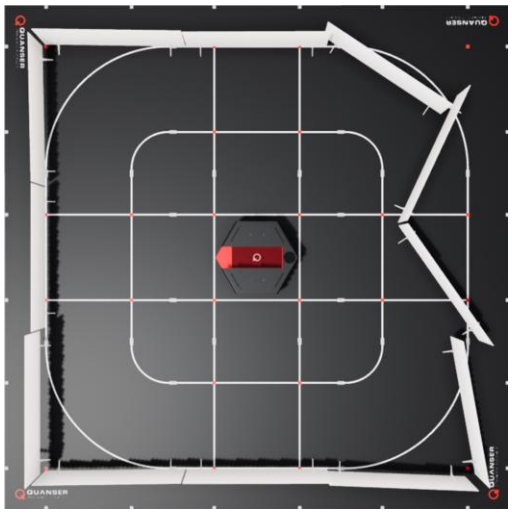


Figure 1: Sample wall layout